FOR A REPRESENTABLE MORPHISM, SEPARATEDNESS IS EQUIVALENT TO THE DIAGONAL BEING A CLOSED IMMERSION, UNIVERSALLY CLOSED OR PROPER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-algebraic-stacks-morphisms/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The criteria for checking a property of a morphism of algebraic stacks along a smooth surjective cover and the base change formula for the diagonal. Invoked inside the counted proof (source lines 475-481) as cross-chapter pointers. Transports each diagonal property between the morphism and its base change along a presentation.
- The equivalence between separatedness and the diagonal being a closed immersion for morphisms of algebraic spaces. Invoked inside the counted proof (source lines 482-485) as a cross-chapter pointer. Gives the equivalences for the presenting morphism of algebraic spaces.
- The base change formula for the diagonal, used a second time. Invoked near the end of the counted proof (source lines 490-493) as a cross-chapter pointer. Identifies the diagonal of an arbitrary base change with a base change of the original diagonal.

Source reference. Original reference categories-lemma-base-change-diagonal: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 476–476. Navigation pointer only: the counted text refers to the Stacks item categories-lemma-base-change-diagonal, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-separated-diagonal-proper: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 484–484. Navigation pointer only: the counted text refers to the Stacks item

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spaces-morphisms-lemma-separated-diagonal-proper, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-check-property-covering: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 479–479. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-check-property-covering, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-section-properties-morphisms: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 248–248. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-section-properties-morphisms, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks representable by algebraic spaces. Then the following are equivalent*

(1) *f is separated,*
(2) *Δ_f is a closed immersion,*
(3) *Δ_f is proper, or*
(4) *Δ_f is universally closed.*

Proof. The statements "f is separated", "Δ_f is a closed immersion", "Δ_f is universally closed", and "Δ_f is proper" refer to the notions defined in Properties of Stacks, Section stacks-properties-section-properties-morphisms. Choose a scheme V and a surjective smooth morphism $V \to \mathcal{Y}$. Set $U = \mathcal{X} \times_{\mathcal{Y}} V$ which is an algebraic space by assumption, and the morphism $U \to \mathcal{X}$ is surjective and smooth. By Categories, Lemma categories-lemma-base-change-diagonal and Properties of Stacks, Lemma stacks-properties-lemma-check-property-covering we see that for any property P (as in that lemma) we have: Δ_f has P if and only if $\Delta_{U/V} : U \to U \times_V U$ has P. Hence the equivalence of (2), (3) and (4) follows from Morphisms of Spaces, Lemma spaces-morphisms-lemma-separated-diagonal-proper applied to $U \to V$. Moreover, if (1) holds, then $U \to V$ is separated and we see that $\Delta_{U/V}$ is a closed immersion, i.e., (2) holds. Finally, assume (2) holds. Let T be a scheme, and $a : T \to \mathcal{Y}$ a morphism. Set $T' = \mathcal{X} \times_{\mathcal{Y}} T$. To prove (1) we have to show that the morphism of algebraic spaces $T' \to T$ is separated. Using Categories, Lemma categories-lemma-base-change-diagonal once more we see that $\Delta_{T'/T}$ is the base change of Δ_f. Hence our assumption (2) implies that $\Delta_{T'/T}$ is a closed immersion, hence $T' \to T$ is separated as desired. □